SEP490 · CAPSTONE PROJECT · REVIEW 2 · BM02

SRS/SDD Self-Review Report – Review 2

1. Run Information

Project code	**FA26SE032**
Project name	Sales Collaborator and Affiliate Network Management System (SCANMS)
Team	Nguyễn Thành Thắng (Leader - SE184251), Nguyễn Đình Tuấn (SE182540), Nguyễn Phú Quý (SE180104), Phan Xuân Thịnh (SE184527)
Supervisor	ThS. Tôn Thất Hoàng Minh
AI tool and model	Antigravity IDE – Claude Opus 4.6, Gemini Flash 3.8
Run date	08/11/2026
Repository tag / commit	https://github.com/thang-nt25/SCANMS-project-capstone-fall26 Main / (commit 99efe6a)
Registration form version used	Version 2.1 (Post-Review 2 Realigned Scope & Immutable Standards)
Version	2.0
Date	08/11/2026

2. Standard Prompt

You are reviewing the SRS and SDD of a software capstone project against its registration form and source code. Report only; do not change any file.

Inputs: the registration form, SRS, SDD (with diagrams), source code and database script in this repository.

Check:
1. **Scope: FRs in the SRS not covered by the registration form, and registration items with no FR.**
2. **Requirements: FRs without an actor or testable acceptance criteria; NFRs without metric, threshold and measurement method.**
3. **Data: entities missing from the Logical ERD; ERD entities or attributes missing from the database script, and tables not in the ERD.**
4. **States: for each main entity, states used in code but missing from its state machine; transitions with no triggering FR; states with no exit.**
5. **Flows: main business flows in activity diagrams with no implementation or no test.**
6. **Architecture: components or third-party services in code but not in the architecture diagram, and the reverse.**
7. **Design vs code: modules, APIs or business rules implemented differently from the SDD.**

Output one table: ID | Category | Location | Finding | Severity (High/Medium/Low) | Evidence. Then list the 5 most important findings.
If something cannot be verified from the repository, say so; do not guess.

3. Findings and Team Response

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-01	Data	docs/database/DATABASE_DESIGN.md & docs/architecture/SAD_DOCUMENT.md vs backend/prisma/schema.prisma	Conceptual & Logical ERD previously depicted open marketplace entities and omitted ProductVariant & OrderItem foreign keys. The data model is now strictly aligned with the Enterprise In-House Multi-Store Collaborator & Affiliate System: full traceability across Customer -> SalesOrder -> OrderItem -> ProductVariant (with floorPrice guard), Store <-> ExclusiveDeal <-> CollaboratorProfile, and CommissionRule / ExclusiveDeal -> Commission.	High	docs/database/DATABASE_DESIGN.md:10-60; backend/prisma/schema.prisma: 13-19, 319-2153	Fixed – Redesigned Conceptual ERD (SVG/PNG) with zero overlapping paths and complete 5-layer relational schema (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).
SR-02	Architecture	docs/architecture/SAD_DOCUMENT.md vs frontend/, Mobile/, backend/	Architecture stack previously referenced Next.js/BullMQ in early documentation. The actual verified production stack is React 19 + Vite (frontend/), React Native / Expo SDK 58 Android Mobile App (Mobile/), NestJS + PostgreSQL + Redis + Socket.IO (backend/), integrated with PayOS/VietQR, GHN Express Sandbox, and Napas 24/7 disbursement.	High	docs/architecture/SAD_DOCUMENT.md: 14-41; frontend/package.json; Mobile/package.json; backend/src/modules/mobile/mobile.controller.ts	Fixed – Updated System Context Diagram and Architectural Design Document to exact deployed stack (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).
SR-03	Scope	Proposed Registration v2.0 vs backend/prisma/schema.prisma & Mobile/src/features/workspace/policy.ts	Legacy marketplace features (open merchant self-registration, standalone shop wallets, public shop vouchers, and Media Hub) were identified as dead code or conceptual drift. All 4 legacy groups are permanently abolished: stores are created exclusively by System Admin, media/files are shared directly via real-time Socket.IO chat, and coupons are strictly limited to KOL Exclusive Deal Coupons.	Medium	frontend/src/routes/AppRoutes.tsx: 203-266; backend/prisma/schema.prisma; Mobile/src/features/workspace/policy.ts:68-71	Fixed – Cleaned obsolete routes, dead UI components, and updated Registration Form v2 & BM01 (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).
SR-04	Flows	Registration v2.0 Deliverables vs Mobile/docs/ROLE_WORKFLOWS_2026-11-03.md	Commission escrow holding duration previously contained a conflict (7 days vs 14 days). The system now strictly standardizes on 14 days (COMMISSION_HOLD_DAYS = 14) matching the 14-day customer return/exchange warranty. Logistics webhook from GHN triggers the 14-day escrow countdown upon successful delivery.	High	Mobile/docs/ROLE_WORKFLOWS_2026-11-03.md: 22, 88; Mobile/src/features/cart/CartScreen.tsx	Fixed – Standardized COMMISSION_HOLD_DAYS = 14 across backend commission constants, DB triggers, and documentation (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-05	Architecture	Registration v2.0 Payouts vs backend/src/modules/payouts/bidv-payout.adapter.ts	Review 2 Council noted the necessity of mobile access for external collaborators and single ownership per workflow. The companion Expo React Native Android app provides real-time chat, sample SLA tracking, vanity coupon management, and two-stage wallet monitoring for Collaborators, while 5 Main Workflows have each been assigned to exactly 1 sole owner.	High	backend/src/modules/payouts/bidv-payout.adapter.ts:15-96; docs/BIDV_SANDBOX.md	Fixed – Mobile companion integrated and 5 Main Workflows strictly mapped to 1 sole owner each (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).
SR-06	Design vs code	Early SDD Commission Spec vs backend/src/modules/commissions/commission-policy.ts & commission.constants.ts	Commission calculation previously had ambiguities regarding additive vs non-stacking deal rates. Strict two-stage non-stacking settlement is now enforced: Stage 1 pays 10% Base + Tier% after 14 days clean delivery; Stage 2 holds the KPI differential max(0, Deal% - (10% + Tier%)) in conditionalDealEscrowBalance until 100% KPI achievement. <=16% approved autonomously by Shop Manager; >16%..25% requires System Admin approval.	Medium	backend/src/modules/commissions/commission-policy.ts:37-47; commission.constants.ts:3; wallets.service.ts:239-307	Fixed – Synchronized commission calculation, wallets ledger, and SDD/SRS specifications to exact two-stage formula (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).
SR-07	States	Early SDD State Diagrams vs backend/prisma/schema.prisma	Dual-path post-sale return resolution: 1-to-1 replacement preserves 100% collaborator commission with carrier dispute claim, whereas customer refund triggers clawback to 0 VND and deducts clean GMV from active Deal KPI and tier promotion progress.	Medium	backend/prisma/schema.prisma: 67-79, 104-135, 213-218; backend/src/modules/returns/return-state.machine.ts	Fixed – Dual-path return state machine and automated clawback posting verified in returns module (Owner: Nguyễn Phú Quý, Completed: 08/11/2026).
SR-08	Design vs code	Early Registration/DB Spec vs backend/src/modules/auth/auth.service.ts & Mobile/src/services/auth/	Authentication & IAM security: Argon2id password hashing, email OTP verification for registration and password reset, and strict role-based access control with maker-checker eKYC verification for external Collaborators.	Medium	backend/src/modules/auth/auth.service.ts: 182-218; Mobile/docs/MOBILE_ARCHITECTURE.md:55-62	Fixed – Unified IAM specification, Argon2id encryption, and maker-checker eKYC pipeline (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).
SR-09	Design vs code	Early Payout Spec vs backend/src/modules/payouts/payouts.service.ts & payout-tax.service.ts	Collaborator withdrawal concurrency and taxation: enforced pessimistic row locking (SELECT ... FOR UPDATE) on wallet balance, mandatory 10% Personal Income Tax (PIT) withholding at source, double-entry FinancePosting audit ledger, and Napas 24/7 disbursement.	Medium	backend/src/modules/payouts/payouts.service.ts:61-115; payout-tax.service.ts:4-22; withdrawal-policy.service.ts:5	Fixed – Concurrency-safe payout processing and 10% PIT withholding verified (Owner: Nguyễn Thành Thắng, Completed: 08/11/2026).

ID	Category	Location	Finding	Severity	Evidence	Team response
SR-10	Requirements	SRS Rate Limiting & AI Spec vs backend/src/core/cache/cache.service.ts & modules/ai/	AI-powered Partner Matchmaking & Fraud Detection: 4-axis collaborator scoring (Category Fit, Engagement, GMV, Quality) and AI Fraud Sentinel for anti-self-referral detection.	Low	backend/src/core/cache/cache.service.ts; backend/src/modules/ai/ai-recommendation.service.ts; ai-fraud.service.ts	Fixed – Deterministic 4-axis AI scoring and rule-based fraud detection verified with RAM caching (Owner: Phan Xuân Thịnh, Completed: 08/11/2026).

4. Top 5 Findings (as reported by the AI)

#	Finding ID	Summary
1	SR-01	In-House Multi-Store Model & Relational ERD Completeness — Standardized enterprise in-house affiliate architecture, eliminating marketplace drift, and completed 5-layer ERD (Customer -> SalesOrder -> OrderItem -> ProductVariant with floorPrice).
2	SR-02	Two-Stage Non-Stacking Commission & 14-Day Escrow — Standardized 14-day return warranty and holding window (COMMISSION_HOLD_DAYS = 14); Stage 1 pays 10%+Tier% after 14 days, Stage 2 holds Deal% - (10%+Tier%) for 100% KPI completion.
3	SR-04	Bi-Directional Exclusive Deal Negotiation via Real-Time Chat — <=16% Shop Manager autonomous approval, >16% to 25% System Admin escalation gate; direct HD media/file exchange replacing deprecated Media Hub.
4	SR-05	Mobile Companion & Cross-Platform Alignment — React Native / Expo SDK 58 Android App connected to shared NestJS API for Collaborator deal chat, sample SLA tracking, vanity coupons, and wallet withdrawal.
5	SR-06	5 Integrated Main Workflows & Sole Ownership Rule — Reorganized 18 features into 5 end-to-end workflows with exactly 1 primary owner each (Thắng WF-01/05, Thịnh WF-02, Tuấn WF-03, Quý WF-04) and balanced 100 test cases.

5. Summary

Severity	Total	Fixed	Kept	Planned
High	4	4	0	0
Medium	5	5	0	0
Low	1	1	0	0
Total	10	10	0	0